

Grounded, Not Automated: A Verification-First LLM Career System

Kayvan Nejabati Zenouz
ZenGrowth (ZENOUZ.ai)
zenouz.ai@gmail.com

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Abstract

AI career tools promise more interviews through resume optimization or unattended application volume, usually without audited outcome denominators. We present ZenGrowth, an audience-of-one, local-first system used in a live senior AI/Data-Science search. Its design constrains generation to a reviewed evidence bank, rejects unallowlisted numeric tokens and detectable named entities, records instrumented model calls, and requires operator approval for every outbound artifact. These mechanisms reduce important fabrication risks; they do not establish sentence-level factual entailment. This v0.2 therefore separates implemented architectural controls from empirical claims that remain to be measured. We pre-register audits of claim-level grounding, repeated-score stability, end-to-end time, and per-application LLM cost. Callback and interview effects are unmeasured. We position the system against prior resume-tailoring pipelines and single-candidate provenance-aware case studies, and use cross-domain productivity experiments only as design analogies, not evidence of job-search effectiveness. The contribution at this stage is a domain-specific operationalization of verification and oversight, a falsifiable evaluation protocol, and a claim-audit checklist for candidate-side LLM systems. Our working hypothesis is deliberately modest: supervised drafting may be useful when errors are cheap to detect, while autonomous submission moves model errors into an irreversible, reputation-bearing action.

1 Introduction

Between 2023 and 2026 the AI job-search category split into two promises. The first is *optimization*: score your resume against a job description and close the keyword gap (Jobscan’s Match Rate, Teal’s Match Score) [11, 16]. The second is *automation*: agents that discover postings and submit applications with minimal human involvement, at volumes of hundreds per week [2, 10]. Both promises are typically supported by testimonials and vendor-reported statistics rather than denominator-complete outcome audits.

ZenGrowth is a system built inside this category but against its grain. It automates everything *except* the send: discovery, explainable scoring, evidence-grounded document generation, interview preparation, and offer evaluation — with a verification path that makes several high-risk forms of fabrication mechanically detectable, and a human-approval boundary that makes every outbound artifact the operator’s deliberate act. It is also an audience-of-one deployment: built by its

operator for a live senior AI/Data-Science search, with workflow events and instrumented model costs logged.

That combination — category-typical capability, category-atypical verification, and an inspectable audit trail — is what makes an honest field note possible. This paper asks three questions. (1) What does our own deployment measurably show (§3)? (2) What is the actual evidence behind the category’s claims (§4)? (3) Where do LLMs genuinely add value in the job search, on the strength of real evidence (§5–§7)?

2 Relation to prior work

Candidate-side LLM systems are no longer absent from the literature. ResumeFlow generates job-specific resumes from a source resume and proposes task-specific alignment and hallucination metrics [19]. The closest prior work is Resume Tailor, a single-candidate case study using a longitudinal career vault, retrieval, provenance tracking, guardrails, and a conditional review loop across nine job descriptions [1]. These papers invalidate any broad claim that personal LLM application systems are undocumented.

ZenGrowth’s narrower proposed delta is *operational and longitudinal*: a live, local-first deployment that connects reviewed evidence, deterministic write-time risk gates, cost and latency telemetry, an application-level action log, and a hard approval boundary. Unlike an ATS-style fit score, its planned evaluation treats unsupported or embellished claims as the primary failure outcome and separately reports downstream job-search outcomes when enough data exist. That delta is a hypothesis until the audits in §3 are complete.

In the broader agent literature, ZenGrowth is not a new general-purpose agent architecture or benchmark. AgencyBench evaluates long-horizon capability over 138 tasks and 32 real-world scenarios [12]. Auditable Agents already formalizes action recoverability, lifecycle coverage, policy checkability, responsibility attribution, and evidence integrity [14]. ZenGrowth should therefore be read as a domain-specific implementation and evaluation opportunity, not as the origin of agent auditability. Recent interview research with 17 experienced developers also identifies a priori control, co-planning, real-time monitoring, and post hoc review as distinct forms of oversight [8]. ZenGrowth currently emphasizes a priori gates and post hoc operator review; it does not yet evaluate the other forms or show that its review is sufficient.

Employer-side work provides important context but not candidate-side outcome evidence. Classical correspondence experiments manipulate perceived identity in otherwise similar applications [6]. More recent audits find demographic sensitivity in embedding retrieval and GPT-based resume assessment [3, 17]; a 2026 audit across newer models reports null or reversed gaps in several settings, showing that direction and magnitude depend on model vintage and protocol [9]. The relevant conclusion is not that all screeners share one stable bias, but that candidate materials enter a model-mediated environment whose behavior requires versioned, task-specific auditing.

3 First-hand evidence: the ZenGrowth deployment

3.1 The system under test

ZenGrowth is a local-first FastAPI/React application with a SQLite store, operated since early 2026 for a live senior AI/DS search. Its pipeline: nightly compliant discovery over public ATS feeds (Greenhouse/Lever JSON; no scraping of job boards) with a dead-man’s-switch heartbeat; an LLM-free relevance precheck; single-call strict-JSON scoring at *temperature* = 0 producing a

per-dimension rationale and a *priority score* — a weighted sum of observable fit dimensions in which the model’s uncalibrated success-probability guess is demoted to a tie-breaking band and application effort is a separate cost axis; and evidence-grounded generation of CVs, cover letters, and application answers.

The verification path is the load-bearing component. Source documents are parsed into claims with provenance; claims auto-verify only when the cited span exists in the source and high-risk tokens match that span, while the rest queue for operator review. Generation receives only verified claims. Write-time gates reject numbers and a conservative subset of named entities found in neither the evidence bank nor the job context. These allowlists catch important fabrication classes but do not prove that every sentence is entailed by its cited evidence; that stronger property is the subject of the planned claim-level audit. Alias resolution merges entity surface variants. A deterministic, zero-LLM-cost JD match report (term coverage, quantified-line count, cliché detection) accompanies every material as a signal, not a target. Interview preparation, debriefs, offer evaluation (web-researched, citation-listed, negotiation-aware), and onboarding/departure planning extend the same pattern: research and drafting with provenance, under operator approval. Nothing is ever sent by the system; drafts carry a never-sent banner until the operator acts.

Instrumented model paths record cost and latency; a configured daily spend ceiling fails closed; and meaningful workflow transitions land in an audit log streamed to the dashboard. The continuously integrated test suite includes deterministic eval gates: hard-fact faithfulness, retrieval recall, entity-resolution pairwise F1, claim-versus-span distortion checks, and a scoring calibration suite (rewording stability bounds, hand-ranked monotonicity, anchor drift alarms) — all CI-enforced with zero LLM spend. Pinning scoring to *temperature* = 0 is a reproducibility control, not a guarantee of identical provider outputs; repeated-score stability is measured separately.

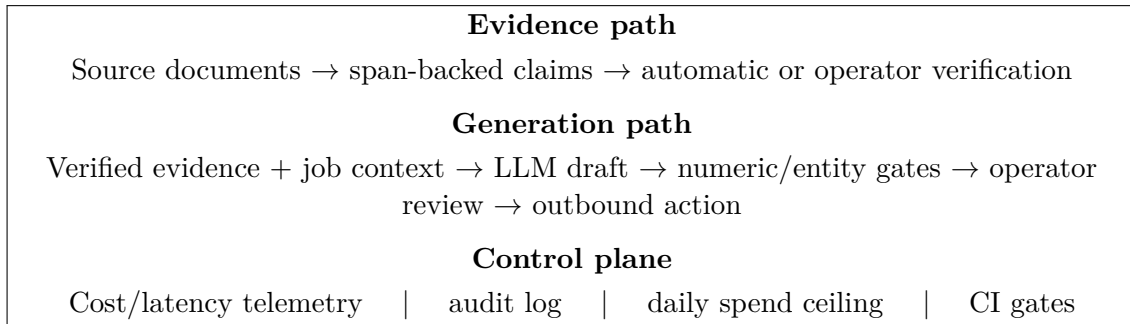


Figure 1: ZenGrowth’s verification-first boundary. Deterministic gates reduce specific risks; the operator remains responsible for semantic verification and all external actions.

3.2 Pre-registered evaluation and current status

Table 1 is a claims scorecard, not a results table. At v0.2 the repository establishes that the listed mechanisms exist and are tested; it does not establish their field rates or downstream value. Measurements will be frozen as dated aggregate artifacts using the public protocol [18] before numerical claims enter the abstract.

Claim	Current verdict	Required measurement
C3/C7	Partially verified: gates exist; semantic fidelity unmeasured	Stratified audit of ≥ 20 generated claims; grounded / unsupported / embellished rates with adjudication rules
C4	Unmeasured: “about two minutes” withdrawn pending timing	≥ 3 end-to-end runs on frozen public postings; wall-clock median and range
C5	Unmeasured: per-call telemetry exists	Per-application cost distribution with operation inclusion rules and currency conversion date
C2/C8	Partially verified: rationale and zero-temperature control exist	3–5 re-scores per frozen input; score and rank variance with model version
C1	Unmeasured: discovery pipeline exists	Archive audit with relevance rubric, sample size, and relevant-role loss rate
C6	Partially verified: local store and documented model egress	Enumerated egress audit: payload class, recipient, trigger, retention, and disclosure
C2b	Unmeasured: usefulness of rank unknown	Rank association with pre-specified pursue decisions; report n and dependence limitations
—	Unmeasured: callback/interview effect	Tracked campaign with complete application denominator and outcome follow-up

Table 1: Claim status and pre-registered measurement plan for v1.

3.3 What the deployment already demonstrates

Some findings do not wait for the quantitative pass because they are architectural facts with test witnesses. The grounding gates fail generation when a detectable, unallowlisted figure or entity appears, and the distortion check demotes extracted claims whose numbers do not match their cited span. Scoring records a rationale and uses a zero-temperature control. A configured daily ceiling blocks further model calls once recorded spend reaches the cap. These are properties of the implementation, not estimates of end-to-end factual accuracy, scoring stability, or user value. The system is also *diagnosable*: logged workflow transitions and instrumented model costs make the planned aggregate analyses possible. That auditability supported the diagnosis of ranking-formula and lifecycle defects during development, while the completeness of the log remains something to test rather than assume.

3.4 Limitations of our evidence

A single operator, a single live search, one configuration lineage, no control arm (no paired untailored applications), and small n throughout; the operator is also the system’s author, so pursue-decisions are not independent of the scoring being evaluated. Downstream outcomes confound with market conditions, seniority, and field. The grounding audit samples materials generated for real applications and therefore cannot be published raw — only aggregate, de-identified rates. These limitations bound the claims: our data will support “measured properties of one audited deployment, with diagnosed mechanisms,” not effect sizes for the category.

4 The AI-career-tool claims landscape

Qualitative synthesis from primary product pages accessed in 2026-07. Product-defined match scores and vendor outcome claims are treated as marketing evidence, not causal estimates.

4.1 Optimization tools: scores without outcome audits

Jobscan recommends a product-defined Match Rate of at least 75%; Teal exposes a 0–100% resume-to-job score and emphasizes keyword and skill alignment [11, 16]. Both publish outcome-oriented claims, but the reviewed pages do not disclose assignment mechanisms, denominator-complete cohorts, attrition, or independent audits. Their scores may be useful editing signals, but they are not calibrated callback probabilities. Classical correspondence studies supply controlled callback evidence for employer discrimination [6]; they do not identify the causal effect of commercial match-score optimization for the same candidate.

4.2 Automation tools: volume as value

JobCopilot advertises up to 50 applications per day and “10X more” interviews; its public page supports the product capability and the existence of the claim, not its causal validity [10]. AIHawk documents an open-source automated application agent; the original repository was archived in May 2026 [2]. Volume is an activity metric. A useful evaluation must also report complete outcome denominators, review time, false submissions, and corrections. Claims about employer detection or platform-wide countermeasures are excluded here because this review did not identify a sufficiently direct primary evidence base.

4.3 The screening environment

Generated applications can enter algorithmic or LLM-mediated screening. Published audits demonstrate demographic sensitivity, but disagree on its direction across models and protocols [3, 9, 17]. For a candidate-side tool this cuts two ways: keyword optimization may exploit model-specific ranking behavior rather than communicate fit, and a fabricated or embellished claim may survive early screening only to create greater risk at interview or verification stages.

5 Cross-domain evidence

No reviewed experiment establishes that LLM-tailored applications improve callback rates. The following cross-domain studies motivate evaluation choices and design hypotheses only; their effect sizes do not transfer to job search.

Bounded assistance. A staggered rollout across 5,179 support agents increased resolutions per hour by 14% on average and 34% for novice and lower-skilled workers, without reducing average customer satisfaction [7]. A separate Copilot experiment found 55.8% faster completion on one specified JavaScript task [15]. These results show that bounded assistance can help in some settings; they do not identify the value of resume tailoring. ZenGrowth’s operator-review boundary is an evidence-informed design choice to be tested, not a result imported from those studies.

Perception is not elapsed time. An early-2025 METR randomized trial of 16 experienced open-source developers found a 19% slowdown while participants estimated after the study that AI had made them 20% faster [4]. METR’s later experiment encountered substantial task and participant selection effects as agentic tools became harder to withhold [5]. The appropriate lesson is methodological: self-reported productivity and task-level timing answer different questions, and both require explicit sampling assumptions.

Supervision has a cost. Agent-usability work frames human supervision time as part of the return-on-investment calculation rather than a free safety layer [13]. This matters in both directions: review can prevent an irreversible submission error, but excessive review can erase any time saving. ZenGrowth therefore needs to measure operator review time and correction burden alongside model latency and spend.

6 A taxonomy of claim failure modes for AI career tools

Failure mode	Mechanism	Category example
Survivorship testimonial ATS folklore	Winners self-report; denominators absent Unverifiable claims about parser behavior drive product design	“I got 5 interviews in a week” “Robots reject resumes without keyword X”
Score–outcome conflation Volume-as-value	Keyword coverage quoted as if it were callbacks Applications sent quoted; interviews per application absent	Match-score marketing Auto-applier dashboards
Perception–reality gap Cost-blind accounting	Users feel more effective; measurement disagrees Subscription, review time, and reputation risk excluded	−19% vs. +20% [4] “Free” bulk applying
Fabrication externality	Embellished claims surface later (interview, reference, background check)	Ungrounded CV generation
Aggregate masking	Averages hide seniority/field segments where the tool fails	Cross-field conversion stats
Capability relabeling	Activity or template completion described as outcome value	Product capability claims

Table 2: Failure modes to test when evaluating AI career-tool claims. The taxonomy is an analytical checklist, not an estimate of category prevalence.

7 Where LLMs genuinely add value in the job search

The architecture follows a testable risk heuristic: assistance is easier to justify when errors are cheap to detect before action, and harder to justify when model output can directly trigger an irreversible outcome. Applied to this system:

Evaluate first: evidence-constrained drafting under operator review; research synthesis with surfaced citations (interview preparation, offer benchmarking); structured extraction checkable against a schema (paste-to-fill intake); deterministic quality signals (match reports, grounding gates) that inform rather than optimize; inspectable audit trails. The evaluation must include correction burden, false reassurance, and review time.

Avoid, or gate hard: unattended submission of applications; keyword optimization as a target rather than a signal; generation without a truth model; volume strategies that spend the candidate’s reputation; treating model confidence as calibrated probability.

In design terms: keep high-risk checks deterministic where possible; let no LLM authority cross the approval boundary; rank on observable signal and demote uncalibrated confidence; freeze baselines before interventions; and use the audit trail to make negative results diagnosable.

8 A checklist for evaluating AI career-tool claims

(1) Outcome data, or testimonial — and is the denominator (total applications) reported? (2) Net of all costs, including subscription, review time, and reputation risk? (3) What is n , and who selected the sample? (4) Stratified by seniority, field, and market, or aggregate? (5) Measured, or self-reported? (6) Who audits the number? (7) Is there a control arm — the same candidate, untailored or unassisted? (8) Would the vendor publish a negative result? (This paper pre-registers its measurements so that item 8 is answered by construction.)

9 Conclusion (provisional at v0.2)

ZenGrowth implements a verification-first alternative to unattended applying: reviewed evidence before generation, deterministic checks for selected high-risk tokens, and operator approval before dispatch. The repository shows that these controls exist and are regression-tested. It does not yet show the rate at which unsupported semantic claims survive, whether repeated scores are stable, whether an application takes two minutes or costs under £1, or whether tailored materials improve callbacks. Those are the pre-registered tests, not conclusions.

The revised novelty claim is correspondingly narrow. Prior systems already generate tailored resumes and use longitudinal retrieval with provenance. ZenGrowth may contribute a longitudinal field account that connects write-time controls, telemetry, correction burden, and downstream outcomes, published with complete denominators. Whether that contribution survives measurement is deliberately left open. At v0.2, the strongest result is a falsifiable protocol and an implemented boundary between model assistance and reputation-bearing action.

Planned revisions. v1: execute and freeze the Table 1 measurements, publish the claim-level adjudication rubric, and report all aggregate results including failures. v2: add tracked-campaign outcome data, an executable replay fixture, and the final venue decision.

Reproducibility

ZenGrowth’s measurement protocol is documented in the repository [18]. The deterministic eval gates that witness architectural controls run in public CI. Dated aggregate baselines and their derivation queries will be published with v1; underlying personal career data will not be published (see Limitations).

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